

Medical Accessibility: AI in Dermatology

A DS 4002 Case Study by Nico Montoya



Being able to detect skin cancer early is an important challenge in medicine today. Despite trained dermatologists being able to identify suspicious lesions with good accuracy, the amount of specialists in the world don't match the demand. This is seen particularly in rural areas as well with patients sometimes having to wait months before being seen for a screening, a time frame that determines the difference between minor procedures and potential life threatening diagnoses.

The medical field has turned over to Data Scientist to help decrease this gap in accessibility. By utilizing "The HAM10000 dataset" provided by Harvard, researchers have been working hard to create models that can categorize common pigmented skin lesions with high precision, but when making a diagnosis, one doesn't only take into account what they can see with their eyes. Demographic factors like age, biological sex, and location of the lesion play important roles as well.

You have been contracted as a Junior Data Scientist for a global health tech giant. Your task is to determine if a "multimodal" approach—combining high-resolution medical imaging with patient metadata—outperforms a model that looks at images alone. As well as decide which parts of the metadata will actually make a difference in the model outcome.

Can machine learning provide a "second opinion" that saves lives? It's time to get in the driver's seat and find out.

References

- [1] P. Tschandl, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." Harvard Dataverse, 2018. doi: 10.7910/DVN/DBW86T.